OVERVIEW OF LINUX SYSTEM

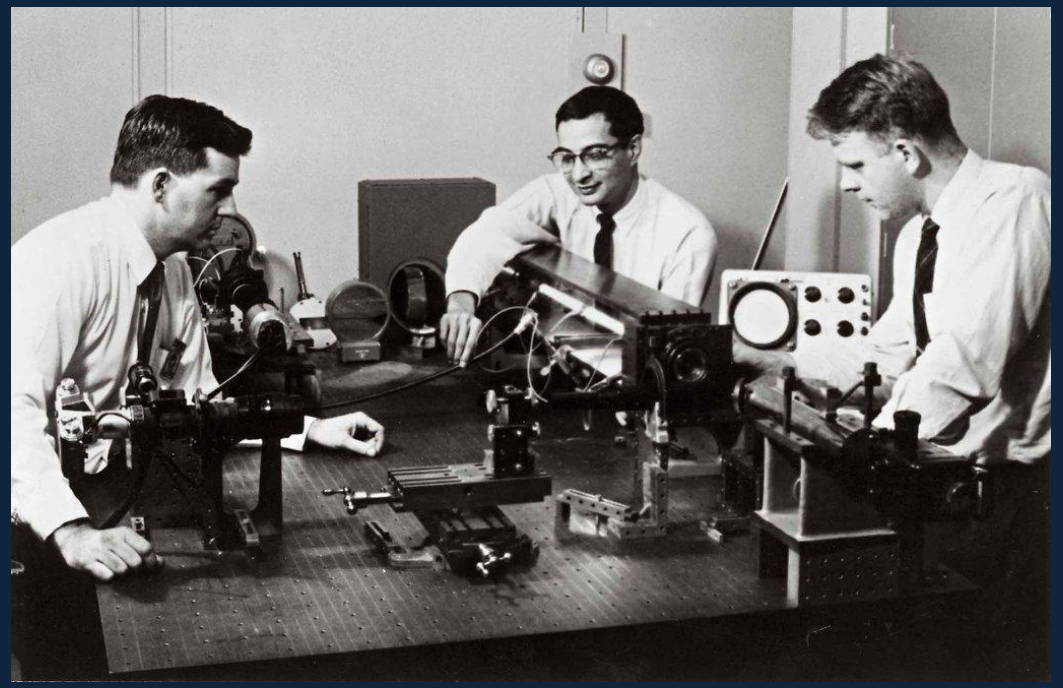
IT 112

Introduction

- Long before Linux the world had Unix.

What is UNIX?

- UNIX is an operating system that originated at Bell Labs (NJ) in 1969.
- UNIX is a trademark but is often used as a generic term to describe “UNIX-like” operating systems



Introduction

- Notable players in the development of the UNIX System are Ken Thompson and Dennis Ritchie.
- The introduction of ‘C’ is also critical to the development of the UNIX system. It was developed to facilitate the creation and rewriting of Unix.

Introduction

- Bell Labs was not able to sell Unix as a product because of antitrust regulations imposed on AT&T at the time.
- Ken Thompson sent media with the Unix code to whoever requested it, often with a note “Love Ken.”
- This “open” approach catch on in later years.

What is Linux?

- A Unix-like operating system initially developed in the early 1990s by Linus Torvald.
- Linus Torvalds is a Finnish-American software engineer.
- Initially developed to run on PC hardware but has been ported to other architectures as well.
- Distributed under a GNU General Public License – “free” software.

Linux OS

- Linux has matured greatly over the past 5 years and has positioned itself as the most flexible UNIX distribution today.
- Notable Linux-Based Operating Systems:
  - Ubuntu, Ubuntu Server, Linux Mint, Fedora(Red Hat), Kali Linux, Raspberry Pi OS, SteamOS (Valve)
- The Linux operating system is in essence a series of files and processes working together to assist the user in completing computational requests.

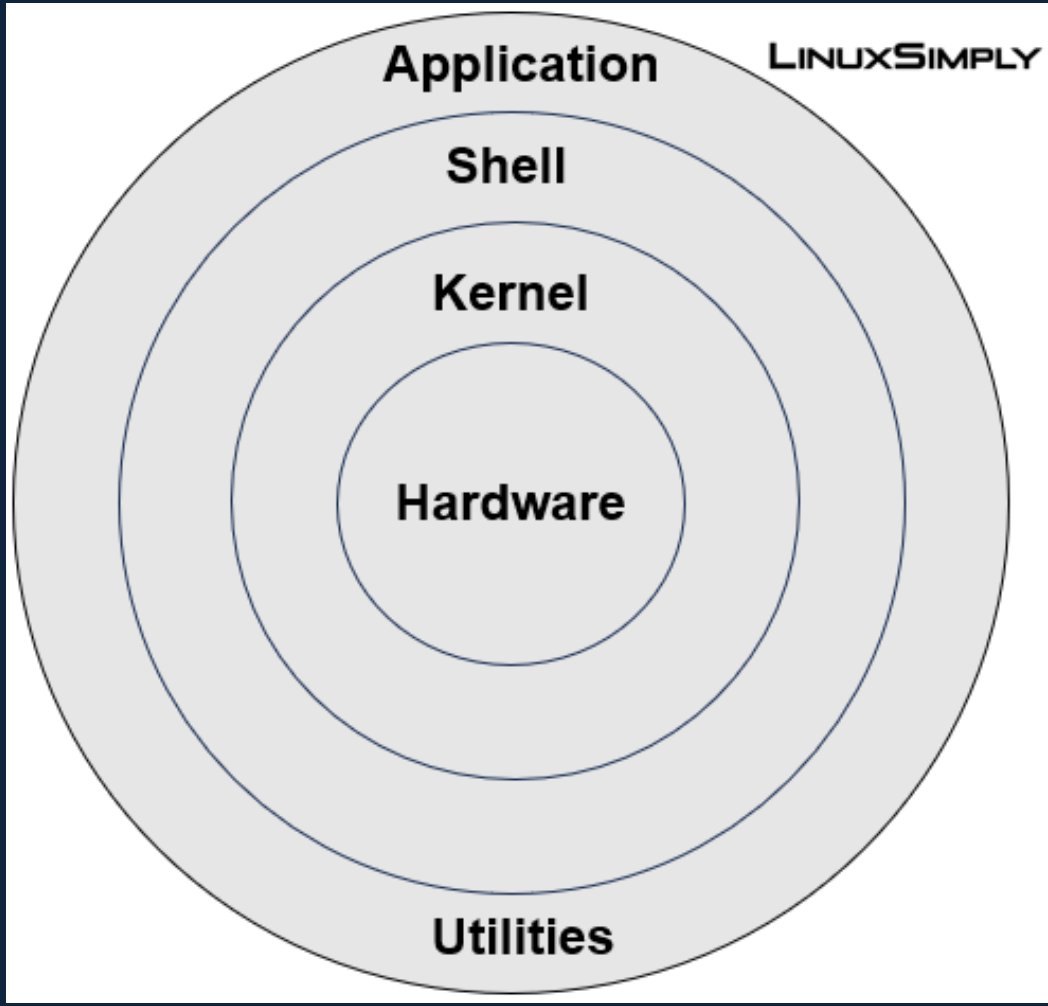
Linux OS System Structure

Layer 1 – Hardware

Layer 2 – Kernel

Layer 3 – Shell Commands

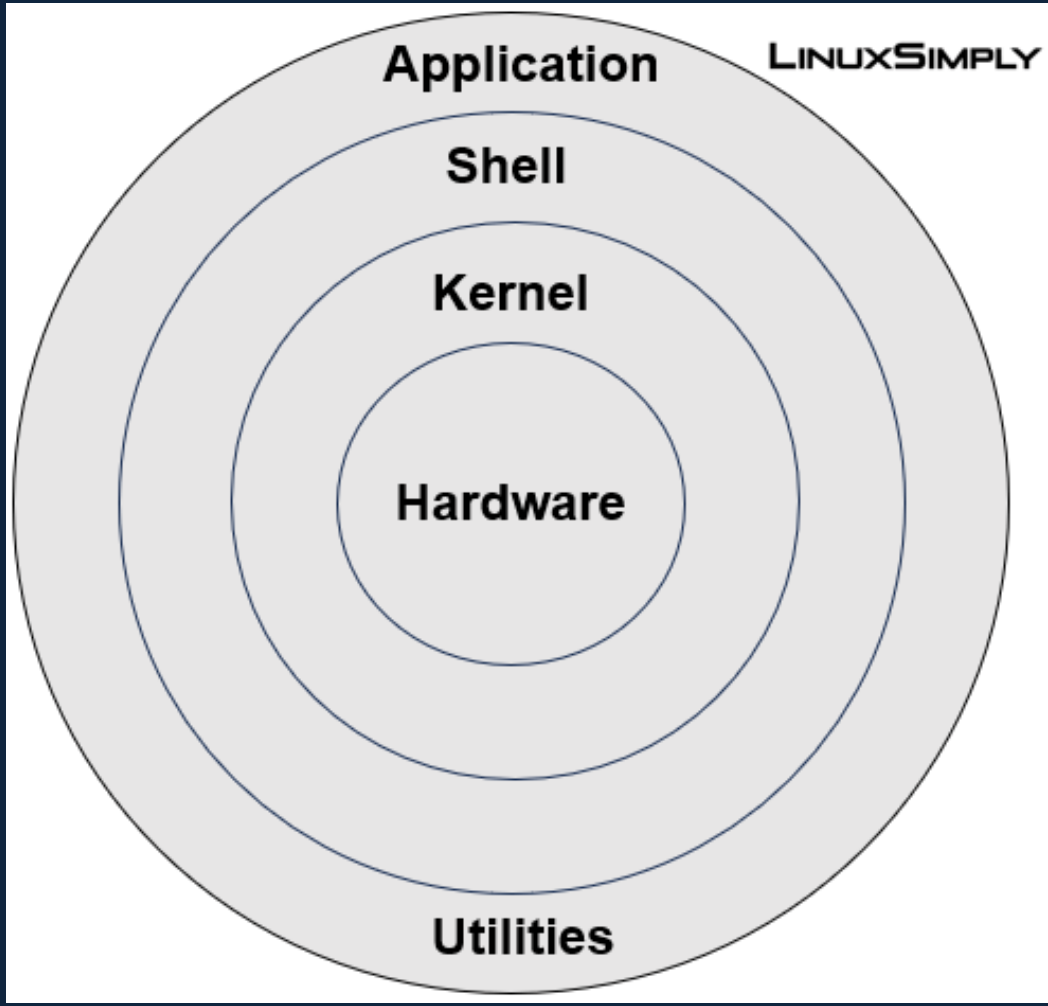
Layer 4 – Application



Linux OS System Structure

Layer 1 – Hardware: It consists of all hardware related information.

- CPU
- RAM
- Input-Output devices

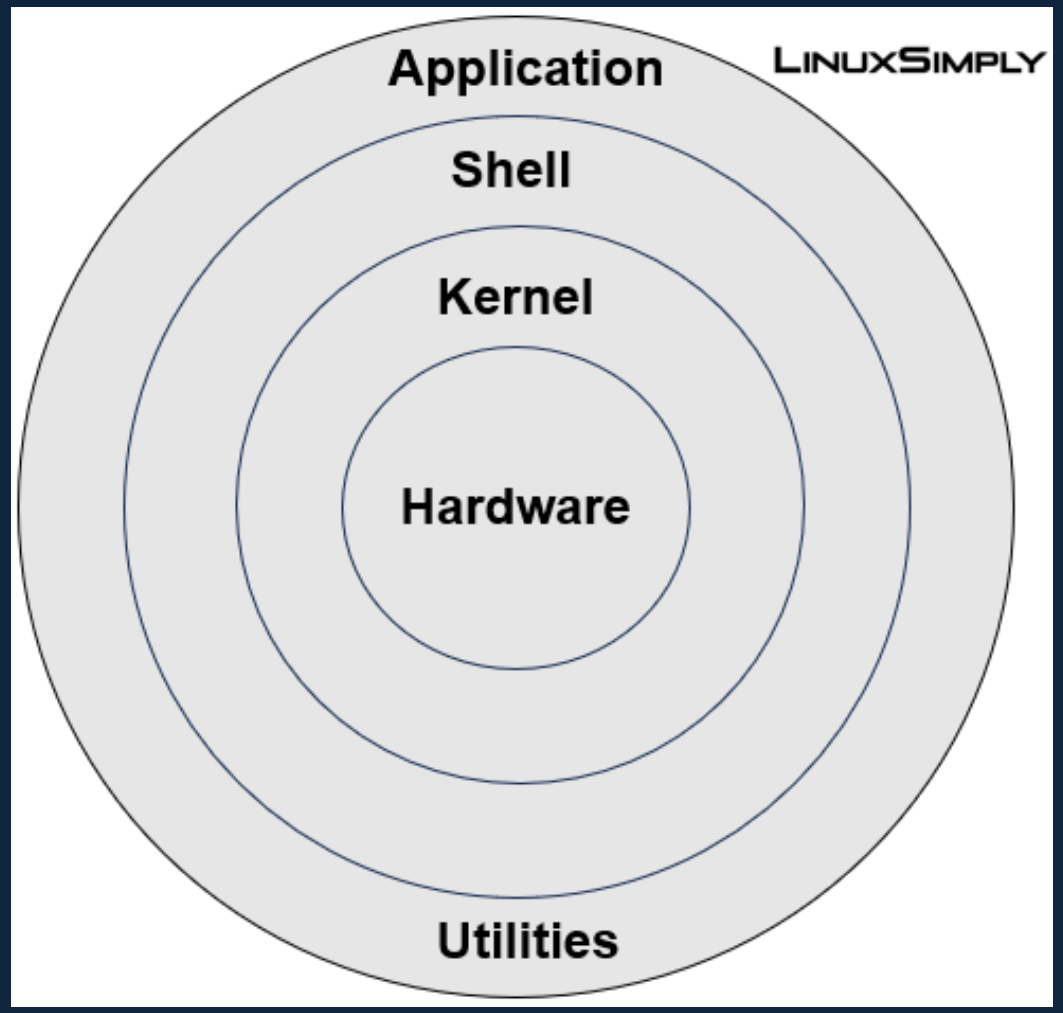


Linux OS System Structure

Layer 2 – Kernel: The core of the operating system and it is the foundation of other software that works on top of it.

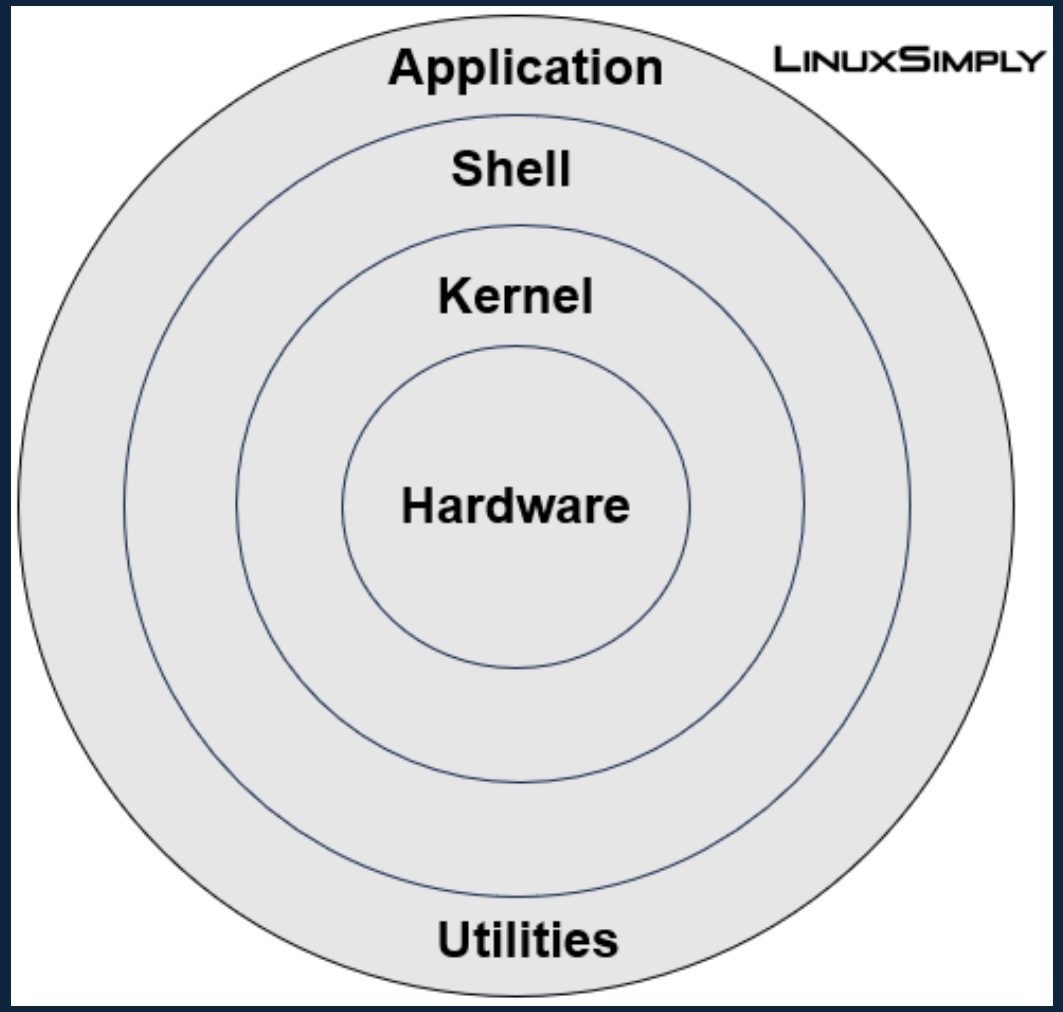
Functions of the kernel:

1. Device management
2. Manage Resources
3. Memory Management
4. System calls
5. Performance optimization



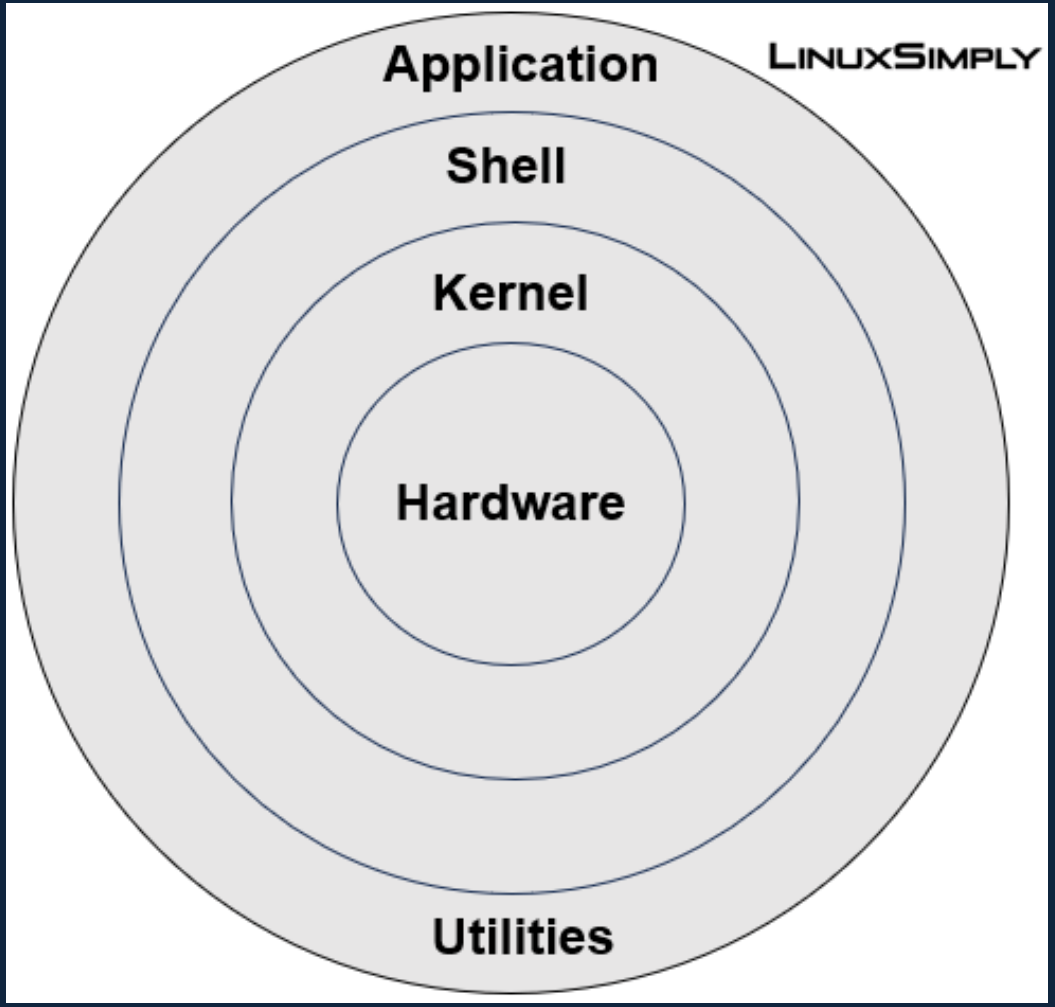
Linux OS System Structure

Layer 3 Shell Commands– Shell is the interface that interacts with humans and processes the commands that are given for the execution. One can call it an interpreter because it takes the command from the keyboard and makes it understandable to the kernel.



Linux OS System Structure

Layer 4 Application – The topmost layer of the Linux architecture consists of applications. These are the software programs that you, as the user, interact with directly. They range from system applications like file managers, text editors, and network managers, to user applications like browsers.



Kernel

- The Linux kernel is the core component of the Linux operating system. It acts as a bridge between hardware and software, managing system resources such as the CPU, memory, and devices.
- It makes the process to wait in the ready queue and execute in consequently to avoid any kind of conflict.
- Types of Kernel:
  - Monolithic Kernel, Micro Kernel, Exo-Kernel, and Hybrid Kerl

Types of Kernel

1. Monolithic Kernel

- The entire OS, including device drivers, memory management, and system calls, runs in a single large kernel space.
- Direct communication with hardware, making it fast but harder to modify.

2. Microkernel

- Only the essential components (process management, memory management, and inter-process communication) run in the kernel.
- Everything else (drivers, file system, etc.) runs in user space, making it modular and secure.

Types of Kernel

3. Hybrid Kernel

- A mix of monolithic and microkernel designs.
- Core services (memory management, process scheduling) run in kernel space, while other services (drivers, file system) run in user space.

4. Exo kernel

- Minimalistic kernel that gives applications direct access to hardware, requiring them to manage resources.
- Exokernel is mostly a research-oriented kernel and is not widely used in commercial systems

Linux System Library

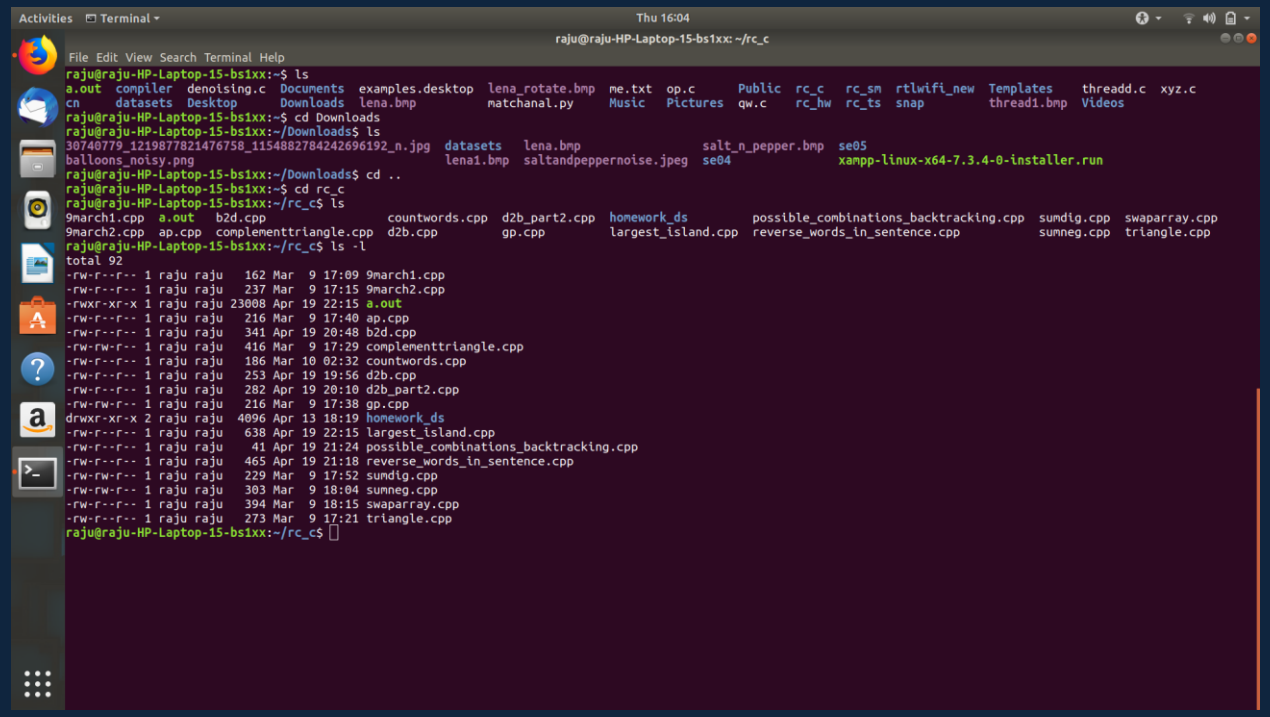
- Linux uses system libraries, also known as shared libraries, to implement various functionalities of the operating system.
- These libraries are the foundation upon which any software can be built.

Linux System Library

- Some of the most common system libraries are:
  - GNU C library: This is the C library that provides the most fundamental system for the interface and execution of C programs. This provides may in-built functions for the execution.
  - libpthread (POSIX Threads): This library plays important role for multithreading in Linux, it allows users for creating and managing multiple threads.
  - libdl (Dynamic Linker): This library is responsible for the loading and linking file at the runtime.
  - libm (Math Library): This library provides user with all kind of mathematical function and their execution.

How is the Linux Operating System Used

- **Servers and Hosting:** Powers web servers, cloud infrastructure, and database management systems.
- **Development:** Used by developers for coding, debugging, and running applications.



How is the Linux Operating System Used

- **Desktop and Personal Use:** Provides secure and customizable desktop environments.
- **Cybersecurity:** Essential for ethical hacking, penetration testing, and security research.

How is the Linux Operating System Used

- **Embedded Systems:** Runs lightweight devices like routers, IoT gadgets, and smart appliances.
- **Supercomputers:** Dominates high-performance computing for scientific research and simulations.

End of Presentation